

Table of contents

\USER

Bochao

Metal_Topup

ACR_EPI_FieldMap

truft_loc_multi_fb
ep2d_se_bw1002_cor_RL
ep2d_se_bw1002_cor_LR
ep2d_se_bw1002_cor_FH
ep2d_se_bw1002_cor_HF
flash_te37
flash_te47
flash_te57
flash_te67
flash_te77
tse_cor_FH
tse_cor_HF

\\USER\Bochao\Metal_Topup\ACR_EPI_FieldMap\trufi_loc_multi_fb

TA: 9.1 s PM: REF Voxel size: 1.7×1.7×8.0 mmPAT: 2 Rel. SNR: 1.00 : tfi

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	On
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	20
Dist. factor	10 %
Position	L4.8 P31.8 F14.5 mm
Orientation	Transversal
Phase enc. dir.	A >> P
Slice group	2
Slices	3
Dist. factor	300 %
Position	L16.4 P43.4 H0.0 mm
Orientation	Sagittal
Phase enc. dir.	A >> P
Slice group	3
Slices	3
Dist. factor	300 %
Position	L15.4 P45.3 H0.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Phase oversampling	0 %
FoV read	400 mm
FoV phase	100.0 %
Slice thickness	8.0 mm
TR	347.70 ms
TE	1.11 ms
Averages	1
Concatenations	26
Filter	Distortion Corr.(2D), Prescan Normalize
Coil elements	BO1,2;SP1-3

Contrast - Common

TR	347.70 ms
TE	1.11 ms
TD	0 ms
Magn. preparation	None
Flip angle	80 deg
Fat suppr.	None
Wrap-up Magn.	Restore

Contrast - Dynamic

Averages	1
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution - Common

FoV read	400 mm
FoV phase	100.0 %
Slice thickness	8.0 mm
Base resolution	240
Phase resolution	66 %
Phase partial Fourier	Off
Trajectory	Cartesian
Interpolation	Off

Resolution - iPAT

PAT mode	GRAPPA
Accel. factor PE	2
Ref. lines PE	24
Reference scan mode	Integrated

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off
Prescan Normalize	On
Unfiltered images	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off
POCS	Off

Geometry - Common

Slice group	1
Slices	20
Dist. factor	10 %
Position	L4.8 P31.8 F14.5 mm
Orientation	Transversal
Phase enc. dir.	A >> P
Slice group	2
Slices	3
Dist. factor	300 %
Position	L16.4 P43.4 H0.0 mm
Orientation	Sagittal
Phase enc. dir.	A >> P
Slice group	3
Slices	3
Dist. factor	300 %
Position	L15.4 P45.3 H0.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
FoV read	400 mm
FoV phase	100.0 %
Slice thickness	8.0 mm
TR	347.70 ms
Multi-slice mode	Sequential
Series	Descending
Concatenations	26

Geometry - AutoAlign

Slice group	1
Position	L4.8 P31.8 F14.5 mm

Geometry - AutoAlign

Orientation	Transversal
Phase enc. dir.	A >> P
Slice group	2
Position	L16.4 P43.4 H0.0 mm
Orientation	Sagittal
Phase enc. dir.	A >> P
Slice group	3
Position	L15.4 P45.3 H0.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Initial Position	L4.8 P31.8 F14.5
Phase	31.8 mm
Read	-4.8 mm
Shift	-14.5 mm
Initial Rotation	0.00 deg
Initial Orientation	Transversal

Geometry - Saturation

Fat suppr.	None
Wrap-up Magn.	Restore
Special sat.	None

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table position	H
Table position	0 mm
Inline Composing	Off

System - Miscellaneous

Positioning mode	REF
Table position	H
Table position	0 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	On - AutoCoilSelect

System - Adjustments

B0 Shim mode	Tune up
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - Tx/Rx

Frequency 1H	23.606969 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	347.70 ms
Concatenations	26
Segments	91

Physio - Cardiac

Tagging	None
Magn. preparation	None
Fat suppr.	None
Dark blood	Off
FoV read	400 mm
FoV phase	100.0 %
Phase resolution	66 %
Cine	Off
Trajectory	Cartesian
Dummy heartbeats	0

Physio - PACE

Resp. control	Off
Concatenations	26

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Save original images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. preparation	None
Contrasts	1
TE	1.11 ms
TR	347.70 ms
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Composing

Inline Composing	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off

Sequence - Part 1

Introduction	Off
Dimension	2D
Reordering	Linear
Asymmetric echo	Weak
Contrasts	1
Optimization	Min. TE

Sequence - Part 1

Multi-slice mode	Sequential
Echo spacing	2.6 ms
Sequence type	Trufi
Bandwidth	1042 Hz/Px

Sequence - Part 2

Define	Shots
Shots per slice	1
Segments	91
Trufi delta freq.	0 Hz
RF pulse type	Fast
Gradient mode	Fast
Excitation	Slice-sel.
Flip angle mode	Constant
Cine	Off

Sequence - Assistant

Mode	Min flip angle
Min flip angle	50 deg
Allowed delay	0 s

\\USER\Bochao\Metal_Topup\ACR_EPI_FieldMap\ep2d_se_bw1002_cor_RL

TA: 0:10 PM: ISO Voxel size: 2.0×2.0×3.0 mmPAT: Off Rel. SNR: 1.00 : epse

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Phase oversampling	0 %
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	4400 ms
TE	149.0 ms
Averages	1
Concatenations	1
Filter	Distortion Corr.(2D)
Coil elements	BO1,2;SP1-3

Contrast - Common

TR	4400 ms
TE	149.0 ms
MTC	Off
Magn. preparation	None
Fat suppr.	None

Contrast - Dynamic

Averages	1
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Delay in TR	0 ms

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
Base resolution	128
Phase resolution	100 %
Phase partial Fourier	Off
Interpolation	Off

Resolution - iPAT

Accel. mode	None
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Resolution - Filter Image

Distortion Corr.	On
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Resolution - Filter Image

Mode	2D
Prescan Normalize	Off
Dynamic Field Corr.	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Geometry - Common

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	4400 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Initial Position	L12.5 P44.3 F17.0
Phase	12.5 mm
Read	17.0 mm
Shift	44.3 mm
Initial Rotation	0.00 deg
Initial Orientation	Coronal

Geometry - Saturation

Fat suppr.	None
Special sat.	None

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table position	F
Table position	17 mm
Inline Composing	Off

System - Miscellaneous

Positioning mode	ISO
Table position	F
Table position	17 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	H >> F
Coil Combine Mode	Adaptive Combine
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	On - AutoCoilSelect

System - Adjustments

B0 Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	256 mm
F >> H	256 mm
A >> P	60 mm
Reset	Off

System - Tx/Rx

Frequency 1H	23.606969 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	3.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	4400 ms
Concatenations	1

Physio - PACE

Resp. control	Off
Concatenations	1

Diff - Neuro

Diffusion mode	Orthogonal
Diff. directions	3
Diffusion Scheme	Monopolar
Diff. weightings	1
b-value	0 s/mm ²
b-value	1
Diff. weighted images	On
Trace weighted images	Off
ADC maps	Off
FA maps	Off
Mosaic	Off
Tensor	Off
Noise level	0

Diff - Body

Diffusion mode	Orthogonal
Diff. directions	3
Diffusion Scheme	Monopolar
Diff. weightings	1
b-value	0 s/mm ²
b-value	1
Diff. weighted images	On
Trace weighted images	Off
ADC maps	Off
Exponential ADC Maps	Off
FA maps	Off
Invert Gray Scale	Off
Calculated Image	Off
b-Value >=	0 s/mm ²
Noise level	0

Diff - Composing

Inline Composing	Off
Distortion Corr.	On
Mode	2D

Sequence - Part 1

Introduction	On
Optimization	Min. TE
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	1.06 ms
Bandwidth	1002 Hz/Px

Sequence - Part 2

EPI factor	128
RF pulse type	Normal
Gradient mode	Fast

Sequence - Special

Panda Tweaks	Off
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\\USER\Bochao\Metal_Topup\ACR_EPI_FieldMap\ep2d_se_bw1002_cor_LR

TA: 0:15 PM: ISO Voxel size: 2.0×2.0×3.0 mmPAT: Off Rel. SNR: 1.00 : epse

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	L >> R
AutoAlign	---
Phase oversampling	0 %
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	4400 ms
TE	149.0 ms
Averages	2
Concatenations	1
Filter	Distortion Corr.(2D)
Coil elements	BO1,2;SP1-3

Contrast - Common

TR	4400 ms
TE	149.0 ms
MTC	Off
Magn. preparation	None
Fat suppr.	None

Contrast - Dynamic

Averages	2
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Delay in TR	0 ms

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
Base resolution	128
Phase resolution	100 %
Phase partial Fourier	Off
Interpolation	Off

Resolution - iPAT

Accel. mode	None
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Resolution - Filter Image

Distortion Corr.	On
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Resolution - Filter Image

Mode	2D
Prescan Normalize	Off
Dynamic Field Corr.	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Geometry - Common

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	L >> R
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	4400 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	L >> R
AutoAlign	---
Initial Position	L12.5 P44.3 F17.0
Phase	-12.5 mm
Read	-17.0 mm
Shift	44.3 mm
Initial Rotation	180.00 deg
Initial Orientation	Coronal

Geometry - Saturation

Fat suppr.	None
Special sat.	None

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table position	F
Table position	17 mm
Inline Composing	Off

System - Miscellaneous

Positioning mode	ISO
Table position	F
Table position	17 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	H >> F
Coil Combine Mode	Adaptive Combine
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	On - AutoCoilSelect

System - Adjustments

B0 Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Rotation	180.00 deg
R >> L	256 mm
F >> H	256 mm
A >> P	60 mm
Reset	Off

System - Tx/Rx

Frequency 1H	23.606969 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	3.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	4400 ms
Concatenations	1

Physio - PACE

Resp. control	Off
Concatenations	1

Diff - Neuro

Diffusion mode	Orthogonal
Diff. directions	3
Diffusion Scheme	Monopolar
Diff. weightings	1
b-value	0 s/mm ²
b-value	2
Diff. weighted images	On
Trace weighted images	Off
ADC maps	Off
FA maps	Off
Mosaic	Off
Tensor	Off
Noise level	0

Diff - Body

Diffusion mode	Orthogonal
Diff. directions	3
Diffusion Scheme	Monopolar
Diff. weightings	1
b-value	0 s/mm ²
b-value	2
Diff. weighted images	On
Trace weighted images	Off
ADC maps	Off
Exponential ADC Maps	Off
FA maps	Off
Invert Gray Scale	Off
Calculated Image	Off
b-Value >=	0 s/mm ²
Noise level	0

Diff - Composing

Inline Composing	Off
Distortion Corr.	On
Mode	2D

Sequence - Part 1

Introduction	On
Optimization	Min. TE
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	1.06 ms
Bandwidth	1002 Hz/Px

Sequence - Part 2

EPI factor	128
RF pulse type	Normal
Gradient mode	Fast

Sequence - Special

Panda Tweaks	Off
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\\USER\Bochao\Metal_Topup\ACR_EPI_FieldMap\ep2d_se_bw1002_cor_FH

TA: 0:15 PM: ISO Voxel size: 2.0×2.0×3.0 mmPAT: Off Rel. SNR: 1.00 : epse

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	F >> H
AutoAlign	---
Phase oversampling	0 %
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	4400 ms
TE	149.0 ms
Averages	2
Concatenations	1
Filter	Distortion Corr.(2D)
Coil elements	BO1,2;SP1-3

Contrast - Common

TR	4400 ms
TE	149.0 ms
MTC	Off
Magn. preparation	None
Fat suppr.	None

Contrast - Dynamic

Averages	2
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Delay in TR	0 ms

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
Base resolution	128
Phase resolution	100 %
Phase partial Fourier	Off
Interpolation	Off

Resolution - iPAT

Accel. mode	None
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Resolution - Filter Image

Distortion Corr.	On
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Resolution - Filter Image

Mode	2D
Prescan Normalize	Off
Dynamic Field Corr.	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Geometry - Common

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	F >> H
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	4400 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	F >> H
AutoAlign	---
Initial Position	L12.5 P44.3 F17.0
Phase	-17.0 mm
Read	12.5 mm
Shift	44.3 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

Geometry - Saturation

Fat suppr.	None
Special sat.	None

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table position	F
Table position	17 mm
Inline Composing	Off

System - Miscellaneous

Positioning mode	ISO
Table position	F
Table position	17 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	H >> F
Coil Combine Mode	Adaptive Combine
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	On - AutoCoilSelect

System - Adjustments

B0 Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	256 mm
R >> L	256 mm
A >> P	60 mm
Reset	Off

System - Tx/Rx

Frequency 1H	23.606969 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	3.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	4400 ms
Concatenations	1

Physio - PACE

Resp. control	Off
Concatenations	1

Diff - Neuro

Diffusion mode	Orthogonal
Diff. directions	3
Diffusion Scheme	Monopolar
Diff. weightings	1
b-value	0 s/mm ²
b-value	2
Diff. weighted images	On
Trace weighted images	Off
ADC maps	Off
FA maps	Off
Mosaic	Off
Tensor	Off
Noise level	0

Diff - Body

Diffusion mode	Orthogonal
Diff. directions	3
Diffusion Scheme	Monopolar
Diff. weightings	1
b-value	0 s/mm ²
b-value	2
Diff. weighted images	On
Trace weighted images	Off
ADC maps	Off
Exponential ADC Maps	Off
FA maps	Off
Invert Gray Scale	Off
Calculated Image	Off
b-Value >=	0 s/mm ²
Noise level	0

Diff - Composing

Inline Composing	Off
Distortion Corr.	On
Mode	2D

Sequence - Part 1

Introduction	On
Optimization	Min. TE
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	1.06 ms
Bandwidth	1002 Hz/Px

Sequence - Part 2

EPI factor	128
RF pulse type	Normal
Gradient mode	Fast

Sequence - Special

Panda Tweaks	Off
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\\USER\Bochao\Metal_Topup\ACR_EPI_FieldMap\ep2d_se_bw1002_cor_HF

TA: 0:15 PM: ISO Voxel size: 2.0×2.0×3.0 mmPAT: Off Rel. SNR: 1.00 : epse

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	H >> F
AutoAlign	---
Phase oversampling	0 %
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	4400 ms
TE	149.0 ms
Averages	2
Concatenations	1
Filter	Distortion Corr.(2D)
Coil elements	BO1,2;SP1-3

Contrast - Common

TR	4400 ms
TE	149.0 ms
MTC	Off
Magn. preparation	None
Fat suppr.	None

Contrast - Dynamic

Averages	2
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Delay in TR	0 ms

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
Base resolution	128
Phase resolution	100 %
Phase partial Fourier	Off
Interpolation	Off

Resolution - iPAT

Accel. mode	None
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Resolution - Filter Image

Distortion Corr.	On
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Resolution - Filter Image

Mode	2D
Prescan Normalize	Off
Dynamic Field Corr.	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Geometry - Common

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	H >> F
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	4400 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	H >> F
AutoAlign	---
Initial Position	L12.5 P44.3 F17.0
Phase	17.0 mm
Read	-12.5 mm
Shift	44.3 mm
Initial Rotation	-90.00 deg
Initial Orientation	Coronal

Geometry - Saturation

Fat suppr.	None
Special sat.	None

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table position	F
Table position	17 mm
Inline Composing	Off

System - Miscellaneous

Positioning mode	ISO
Table position	F
Table position	17 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	H >> F
Coil Combine Mode	Adaptive Combine
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	On - AutoCoilSelect

System - Adjustments

B0 Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Rotation	-90.00 deg
F >> H	256 mm
R >> L	256 mm
A >> P	60 mm
Reset	Off

System - Tx/Rx

Frequency 1H	23.606969 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	3.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	4400 ms
Concatenations	1

Physio - PACE

Resp. control	Off
Concatenations	1

Diff - Neuro

Diffusion mode	Orthogonal
Diff. directions	3
Diffusion Scheme	Monopolar
Diff. weightings	1
b-value	0 s/mm ²
b-value	2
Diff. weighted images	On
Trace weighted images	Off
ADC maps	Off
FA maps	Off
Mosaic	Off
Tensor	Off
Noise level	0

Diff - Body

Diffusion mode	Orthogonal
Diff. directions	3
Diffusion Scheme	Monopolar
Diff. weightings	1
b-value	0 s/mm ²
b-value	2
Diff. weighted images	On
Trace weighted images	Off
ADC maps	Off
Exponential ADC Maps	Off
FA maps	Off
Invert Gray Scale	Off
Calculated Image	Off
b-Value >=	0 s/mm ²
Noise level	0

Diff - Composing

Inline Composing	Off
Distortion Corr.	On
Mode	2D

Sequence - Part 1

Introduction	On
Optimization	Min. TE
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	1.06 ms
Bandwidth	1002 Hz/Px

Sequence - Part 2

EPI factor	128
RF pulse type	Normal
Gradient mode	Fast

Sequence - Special

Panda Tweaks	Off
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\\USER\Bochao\Metal_Topup\ACR_EPI_FieldMap\flash_te37

TA: 1:20 PM: ISO Voxel size: 2.0×2.0×3.0 mmPAT: Off Rel. SNR: 1.00 : fl

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off
Prescan Normalize	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Routine

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Phase oversampling	0 %
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	205.0 ms
TE	3.70 ms
Averages	3
Concatenations	1
Filter	Distortion Corr.(2D)
Coil elements	BO1,2;SP1-3

Geometry - Common

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	205.0 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Initial Position	L12.5 P44.3 F17.0
Phase	12.5 mm
Read	17.0 mm
Shift	44.3 mm
Initial Rotation	0.00 deg
Initial Orientation	Coronal

Contrast - Common

TR	205.0 ms
TE	3.70 ms
MTC	Off
Magn. preparation	None
Flip angle	25 deg
Fat suppr.	None
Water suppr.	None
SWI	Off

Geometry - Saturation

Saturation mode	Standard
Fat suppr.	None
Water suppr.	None
Special sat.	None

Contrast - Dynamic

Averages	3
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table position	F
Table position	17 mm
Inline Composing	Off

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
Base resolution	128
Phase resolution	100 %
Phase partial Fourier	Off
Interpolation	Off

Geometry - Tim CT

Tim CT mode	Off
Slices	20
Slice thickness	3.0 mm
Dist. factor	0 %
FoV read	256 mm
FoV phase	100.0 %
Segments	1

Resolution - iPAT

PAT mode	None
----------	------

System - Miscellaneous

Positioning mode	ISO
Table position	F
Table position	17 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	H >> F
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	256 mm
F >> H	256 mm
A >> P	60 mm
Reset	Off

System - Tx/Rx

Frequency 1H	23.606969 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	205.0 ms
Concatenations	1
Segments	1

Physio - Cardiac

Tagging	None
Magn. preparation	None
Fat suppr.	None
Dark blood	Off
FoV read	256 mm
FoV phase	100.0 %
Phase resolution	100 %

Physio - PACE

Resp. control	Off
Concatenations	1

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Liver registration	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Soft Tissue

Wash - In	Off
Wash - Out	Off
TTP	Off
PEI	Off
MIP - time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off

Inline - MapIt

Save original images	On
MapIt	None
Flip angle	25 deg
Measurements	1
Contrasts	1
TR	205.0 ms
TE	3.70 ms

Sequence - Part 1

Introduction	On
Dimension	2D
Phase stabilisation	Off
Asymmetric echo	Off
Contrasts	1
Flow comp.	No
Multi-slice mode	Interleaved
Bandwidth	590 Hz/Px

Sequence - Part 2

Segments	1
Acoustic noise reduction	None
RF pulse type	Normal
Gradient mode	Fast
Excitation	Slice-sel.
RF spoiling	On

Sequence - Special

Panda Tweaks	Off
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Sequence - Assistant

Mode	Off
------	-----

\\USER\Bochao\Metal_Topup\ACR_EPI_FieldMap\flash_te47

TA: 1:20 PM: ISO Voxel size: 2.0×2.0×3.0 mmPAT: Off Rel. SNR: 1.00 : fl

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off
Prescan Normalize	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Routine

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Phase oversampling	0 %
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	205.0 ms
TE	4.70 ms
Averages	3
Concatenations	1
Filter	Distortion Corr.(2D)
Coil elements	BO1,2;SP1-3

Geometry - Common

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	205.0 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Initial Position	L12.5 P44.3 F17.0
Phase	12.5 mm
Read	17.0 mm
Shift	44.3 mm
Initial Rotation	0.00 deg
Initial Orientation	Coronal

Contrast - Common

TR	205.0 ms
TE	4.70 ms
MTC	Off
Magn. preparation	None
Flip angle	25 deg
Fat suppr.	None
Water suppr.	None
SWI	Off

Geometry - Saturation

Saturation mode	Standard
Fat suppr.	None
Water suppr.	None
Special sat.	None

Contrast - Dynamic

Averages	3
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table position	F
Table position	17 mm
Inline Composing	Off

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
Base resolution	128
Phase resolution	100 %
Phase partial Fourier	Off
Interpolation	Off

Geometry - Tim CT

Tim CT mode	Off
Slices	20
Slice thickness	3.0 mm
Dist. factor	0 %
FoV read	256 mm
FoV phase	100.0 %
Segments	1

Resolution - iPAT

PAT mode	None
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System - Miscellaneous

Positioning mode	ISO
Table position	F
Table position	17 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	H >> F
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	256 mm
F >> H	256 mm
A >> P	60 mm
Reset	Off

System - Tx/Rx

Frequency 1H	23.606969 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	205.0 ms
Concatenations	1
Segments	1

Physio - Cardiac

Tagging	None
Magn. preparation	None
Fat suppr.	None
Dark blood	Off
FoV read	256 mm
FoV phase	100.0 %
Phase resolution	100 %

Physio - PACE

Resp. control	Off
Concatenations	1

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Liver registration	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Soft Tissue

Wash - In	Off
Wash - Out	Off
TTP	Off
PEI	Off
MIP - time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off

Inline - MapIt

Save original images	On
MapIt	None
Flip angle	25 deg
Measurements	1
Contrasts	1
TR	205.0 ms
TE	4.70 ms

Sequence - Part 1

Introduction	On
Dimension	2D
Phase stabilisation	Off
Asymmetric echo	Off
Contrasts	1
Flow comp.	No
Multi-slice mode	Interleaved
Bandwidth	590 Hz/Px

Sequence - Part 2

Segments	1
Acoustic noise reduction	None
RF pulse type	Normal
Gradient mode	Fast
Excitation	Slice-sel.
RF spoiling	On

Sequence - Special

Panda Tweaks	Off
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Sequence - Assistant

Mode	Off
------	-----

\\USER\Bochao\Metal_Topup\ACR_EPI_FieldMap\flash_te57

TA: 1:20 PM: ISO Voxel size: 2.0×2.0×3.0 mmPAT: Off Rel. SNR: 1.00 : fl

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off
Prescan Normalize	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Routine

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Phase oversampling	0 %
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	205.0 ms
TE	5.70 ms
Averages	3
Concatenations	1
Filter	Distortion Corr.(2D)
Coil elements	BO1,2;SP1-3

Geometry - Common

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	205.0 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Initial Position	L12.5 P44.3 F17.0
Phase	12.5 mm
Read	17.0 mm
Shift	44.3 mm
Initial Rotation	0.00 deg
Initial Orientation	Coronal

Contrast - Common

TR	205.0 ms
TE	5.70 ms
MTC	Off
Magn. preparation	None
Flip angle	25 deg
Fat suppr.	None
Water suppr.	None
SWI	Off

Geometry - Saturation

Saturation mode	Standard
Fat suppr.	None
Water suppr.	None
Special sat.	None

Contrast - Dynamic

Averages	3
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table position	F
Table position	17 mm
Inline Composing	Off

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
Base resolution	128
Phase resolution	100 %
Phase partial Fourier	Off
Interpolation	Off

Geometry - Tim CT

Tim CT mode	Off
Slices	20
Slice thickness	3.0 mm
Dist. factor	0 %
FoV read	256 mm
FoV phase	100.0 %
Segments	1

Resolution - iPAT

PAT mode	None
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System - Miscellaneous

Positioning mode	ISO
Table position	F
Table position	17 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	H >> F
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	256 mm
F >> H	256 mm
A >> P	60 mm
Reset	Off

System - Tx/Rx

Frequency 1H	23.606969 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	205.0 ms
Concatenations	1
Segments	1

Physio - Cardiac

Tagging	None
Magn. preparation	None
Fat suppr.	None
Dark blood	Off
FoV read	256 mm
FoV phase	100.0 %
Phase resolution	100 %

Physio - PACE

Resp. control	Off
Concatenations	1

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Liver registration	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Soft Tissue

Wash - In	Off
Wash - Out	Off
TTP	Off
PEI	Off
MIP - time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off

Inline - MapIt

Save original images	On
MapIt	None
Flip angle	25 deg
Measurements	1
Contrasts	1
TR	205.0 ms
TE	5.70 ms

Sequence - Part 1

Introduction	On
Dimension	2D
Phase stabilisation	Off
Asymmetric echo	Off
Contrasts	1
Flow comp.	No
Multi-slice mode	Interleaved
Bandwidth	590 Hz/Px

Sequence - Part 2

Segments	1
Acoustic noise reduction	None
RF pulse type	Normal
Gradient mode	Fast
Excitation	Slice-sel.
RF spoiling	On

Sequence - Special

Panda Tweaks	Off
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Sequence - Assistant

Mode	Off
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\\USER\Bochao\Metal_Topup\ACR_EPI_FieldMap\flash_te67

TA: 1:20 PM: ISO Voxel size: 2.0×2.0×3.0 mmPAT: Off Rel. SNR: 1.00 : fl

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off
Prescan Normalize	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Routine

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Phase oversampling	0 %
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	205.0 ms
TE	6.70 ms
Averages	3
Concatenations	1
Filter	Distortion Corr.(2D)
Coil elements	BO1,2;SP1-3

Geometry - Common

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	205.0 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Initial Position	L12.5 P44.3 F17.0
Phase	12.5 mm
Read	17.0 mm
Shift	44.3 mm
Initial Rotation	0.00 deg
Initial Orientation	Coronal

Contrast - Common

TR	205.0 ms
TE	6.70 ms
MTC	Off
Magn. preparation	None
Flip angle	25 deg
Fat suppr.	None
Water suppr.	None
SWI	Off

Geometry - Saturation

Saturation mode	Standard
Fat suppr.	None
Water suppr.	None
Special sat.	None

Contrast - Dynamic

Averages	3
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table position	F
Table position	17 mm
Inline Composing	Off

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
Base resolution	128
Phase resolution	100 %
Phase partial Fourier	Off
Interpolation	Off

Geometry - Tim CT

Tim CT mode	Off
Slices	20
Slice thickness	3.0 mm
Dist. factor	0 %
FoV read	256 mm
FoV phase	100.0 %
Segments	1

Resolution - iPAT

PAT mode	None
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System - Miscellaneous

Positioning mode	ISO
Table position	F
Table position	17 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	H >> F
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	256 mm
F >> H	256 mm
A >> P	60 mm
Reset	Off

System - Tx/Rx

Frequency 1H	23.606969 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	205.0 ms
Concatenations	1
Segments	1

Physio - Cardiac

Tagging	None
Magn. preparation	None
Fat suppr.	None
Dark blood	Off
FoV read	256 mm
FoV phase	100.0 %
Phase resolution	100 %

Physio - PACE

Resp. control	Off
Concatenations	1

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Liver registration	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Soft Tissue

Wash - In	Off
Wash - Out	Off
TTP	Off
PEI	Off
MIP - time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off

Inline - MapIt

Save original images	On
MapIt	None
Flip angle	25 deg
Measurements	1
Contrasts	1
TR	205.0 ms
TE	6.70 ms

Sequence - Part 1

Introduction	On
Dimension	2D
Phase stabilisation	Off
Asymmetric echo	Off
Contrasts	1
Flow comp.	No
Multi-slice mode	Interleaved
Bandwidth	590 Hz/Px

Sequence - Part 2

Segments	1
Acoustic noise reduction	None
RF pulse type	Normal
Gradient mode	Fast
Excitation	Slice-sel.
RF spoiling	On

Sequence - Special

Panda Tweaks	Off
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Sequence - Assistant

Mode	Off
------	-----

\\USER\Bochao\Metal_Topup\ACR_EPI_FieldMap\flash_te77

TA: 1:20 PM: ISO Voxel size: 2.0×2.0×3.0 mmPAT: Off Rel. SNR: 1.00 : fl

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Resolution - Filter Image

Image Filter	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off
Prescan Normalize	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Routine

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Phase oversampling	0 %
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	205.0 ms
TE	7.70 ms
Averages	3
Concatenations	1
Filter	Distortion Corr.(2D)
Coil elements	BO1,2;SP1-3

Geometry - Common

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	205.0 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	R >> L
AutoAlign	---
Initial Position	L12.5 P44.3 F17.0
Phase	12.5 mm
Read	17.0 mm
Shift	44.3 mm
Initial Rotation	0.00 deg
Initial Orientation	Coronal

Contrast - Common

TR	205.0 ms
TE	7.70 ms
MTC	Off
Magn. preparation	None
Flip angle	25 deg
Fat suppr.	None
Water suppr.	None
SWI	Off

Contrast - Dynamic

Averages	3
Averaging mode	Short term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
Base resolution	128
Phase resolution	100 %
Phase partial Fourier	Off
Interpolation	Off

Resolution - iPAT

PAT mode	None
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Geometry - Saturation

Saturation mode	Standard
Fat suppr.	None
Water suppr.	None
Special sat.	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table position	F
Table position	17 mm
Inline Composing	Off

Geometry - Tim CT

Tim CT mode	Off
Slices	20
Slice thickness	3.0 mm
Dist. factor	0 %
FoV read	256 mm
FoV phase	100.0 %
Segments	1

System - Miscellaneous

Positioning mode	ISO
Table position	F
Table position	17 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	H >> F
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	Default

System - Adjustments

B0 Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Rotation	0.00 deg
R >> L	256 mm
F >> H	256 mm
A >> P	60 mm
Reset	Off

System - Tx/Rx

Frequency 1H	23.606969 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	205.0 ms
Concatenations	1
Segments	1

Physio - Cardiac

Tagging	None
Magn. preparation	None
Fat suppr.	None
Dark blood	Off
FoV read	256 mm
FoV phase	100.0 %
Phase resolution	100 %

Physio - PACE

Resp. control	Off
Concatenations	1

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Liver registration	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Soft Tissue

Wash - In	Off
Wash - Out	Off
TTP	Off
PEI	Off
MIP - time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off

Inline - MapIt

Save original images	On
MapIt	None
Flip angle	25 deg
Measurements	1
Contrasts	1
TR	205.0 ms
TE	7.70 ms

Sequence - Part 1

Introduction	On
Dimension	2D
Phase stabilisation	Off
Asymmetric echo	Off
Contrasts	1
Flow comp.	No
Multi-slice mode	Interleaved
Bandwidth	590 Hz/Px

Sequence - Part 2

Segments	1
Acoustic noise reduction	None
RF pulse type	Normal
Gradient mode	Fast
Excitation	Slice-sel.
RF spoiling	On

Sequence - Special

Panda Tweaks	Off
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Sequence - Assistant

Mode	Off
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\\USER\Bochao\Metal_Topup\ACR_EPI_FieldMap\tse_cor_FH

TA: 0:52 PM: ISO Voxel size: 2.0×2.0×3.0 mmPAT: Off Rel. SNR: 1.00 : tseR

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	F >> H
AutoAlign	---
Phase oversampling	0 %
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	5610.0 ms
TE	114 ms
Averages	1
Concatenations	1
Filter	Distortion Corr.(2D)
Coil elements	BO1,2;SP1-3

Contrast - Common

TR	5610.0 ms
TE	114 ms
MTC	Off
Magn. preparation	None
Flip angle	160 deg
Fat suppr.	None
Water suppr.	None
Restore magn.	On

Contrast - Dynamic

Averages	1
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
Base resolution	128
Phase resolution	100 %
Phase partial Fourier	Off
Trajectory	Cartesian
Interpolation	Off

Resolution - iPAT

PAT mode	None
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Resolution - Filter Image

Image Filter	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off
Prescan Normalize	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Geometry - Common

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	F >> H
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	5610.0 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	F >> H
AutoAlign	---
Initial Position	L12.5 P44.3 F17.0
Phase	-17.0 mm
Read	12.5 mm
Shift	44.3 mm
Initial Rotation	90.00 deg
Initial Orientation	Coronal

Geometry - Saturation

Fat suppr.	None
Water suppr.	None
Restore magn.	On
Special sat.	None

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table position	F
Table position	17 mm
Inline Composing	Off

Geometry - Tim CT

Tim CT mode	Off
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Geometry - Tim CT

Slices	20
Slice thickness	3.0 mm
Dist. factor	0 %
FoV read	256 mm
FoV phase	100.0 %

System - Miscellaneous

Positioning mode	ISO
Table position	F
Table position	17 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	H >> F
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	On - AutoCoilSelect

System - Adjustments

B0 Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Rotation	90.00 deg
F >> H	256 mm
R >> L	256 mm
A >> P	60 mm
Reset	Off

System - Tx/Rx

Frequency 1H	23.606969 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	5610.0 ms
Concatenations	1

Physio - Cardiac

Magn. preparation	None
Fat suppr.	None
Dark blood	Off
FoV read	256 mm
FoV phase	100.0 %
Phase resolution	100 %
Trajectory	Cartesian

Physio - PACE

Resp. control	Off
Concatenations	1

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Composing

Inline Composing	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off

Sequence - Part 1

Introduction	On
Dimension	2D
Compensate T2 decay	Off
Reduce Motion Sens.	Off
Contrasts	1
Flow comp.	No
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	12.7 ms
Bandwidth	120 Hz/Px

Sequence - Part 2

Define	Turbo factor
Echo trains per slice	8
Phase correction	Automatic
Acoustic noise reduction	None
RF pulse type	Normal
Gradient mode	Normal
WARP	Off
Red. EC sensitivity	Off
Turbo factor	17

Sequence - Special

Panda Tweaks	Off
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Sequence - Assistant

Mode	TR
Max. TR	5000.0 ms
Allowed delay	30 s

\\USER\Bochao\Metal_Topup\ACR_EPI_FieldMap\tse_cor_HF

TA: 0:52 PM: ISO Voxel size: 2.0×2.0×3.0 mmPAT: Off Rel. SNR: 1.00 : tseR

Properties

Prio recon	Off
Load images to viewer	On
Inline movie	Off
Auto store images	On
Load images to stamp segments	Off
Load images to graphic segments	Off
Auto open inline display	Off
Auto close inline display	Off
Start measurement without further preparation	Off
Wait for user to start	Off
Start measurements	Single measurement

Routine

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	H >> F
AutoAlign	---
Phase oversampling	0 %
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	5610.0 ms
TE	114 ms
Averages	1
Concatenations	1
Filter	Distortion Corr.(2D)
Coil elements	BO1,2;SP1-3

Contrast - Common

TR	5610.0 ms
TE	114 ms
MTC	Off
Magn. preparation	None
Flip angle	160 deg
Fat suppr.	None
Water suppr.	None
Restore magn.	On

Contrast - Dynamic

Averages	1
Averaging mode	Long term
Reconstruction	Magnitude
Measurements	1
Multiple series	Each measurement

Resolution - Common

FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
Base resolution	128
Phase resolution	100 %
Phase partial Fourier	Off
Trajectory	Cartesian
Interpolation	Off

Resolution - iPAT

PAT mode	None
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Resolution - Filter Image

Image Filter	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off
Prescan Normalize	Off
Normalize	Off
B1 filter	Off

Resolution - Filter Rawdata

Raw filter	Off
Elliptical filter	Off

Geometry - Common

Slice group	1
Slices	20
Dist. factor	0 %
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	H >> F
FoV read	256 mm
FoV phase	100.0 %
Slice thickness	3.0 mm
TR	5610.0 ms
Multi-slice mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice group	1
Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Phase enc. dir.	H >> F
AutoAlign	---
Initial Position	L12.5 P44.3 F17.0
Phase	17.0 mm
Read	-12.5 mm
Shift	44.3 mm
Initial Rotation	-90.00 deg
Initial Orientation	Coronal

Geometry - Saturation

Fat suppr.	None
Water suppr.	None
Restore magn.	On
Special sat.	None

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table position	F
Table position	17 mm
Inline Composing	Off

Geometry - Tim CT

Tim CT mode	Off
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Geometry - Tim CT

Slices	20
Slice thickness	3.0 mm
Dist. factor	0 %
FoV read	256 mm
FoV phase	100.0 %

System - Miscellaneous

Positioning mode	ISO
Table position	F
Table position	17 mm
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	H >> F
Coil Combine Mode	Adaptive Combine
Save uncombined	Off
Matrix Optimization	Off
AutoAlign	---
Coil Select Mode	On - AutoCoilSelect

System - Adjustments

B0 Shim mode	Standard
Adjust with body coil	Off
Confirm freq. adjustment	Off
Assume Dominant Fat	Off
Assume Silicone	Off
Adjustment Tolerance	Auto

System - Adjust Volume

Position	L12.5 P44.3 F17.0 mm
Orientation	Coronal
Rotation	-90.00 deg
F >> H	256 mm
R >> L	256 mm
A >> P	60 mm
Reset	Off

System - Tx/Rx

Frequency 1H	23.606969 MHz
Correction factor	1
Gain	High
Img. Scale Cor.	1.000
Reset	Off
? Ref. amplitude 1H	0.000 V

Physio - Signal1

1st Signal/Mode	None
TR	5610.0 ms
Concatenations	1

Physio - Cardiac

Magn. preparation	None
Fat suppr.	None
Dark blood	Off
FoV read	256 mm
FoV phase	100.0 %
Phase resolution	100 %
Trajectory	Cartesian

Physio - PACE

Resp. control	Off
Concatenations	1

Inline - Common

Subtract	Off
Measurements	1
StdDev	Off
Save original images	On

Inline - MIP

MIP-Sag	Off
MIP-Cor	Off
MIP-Tra	Off
MIP-Time	Off
Save original images	On

Inline - Composing

Inline Composing	Off
Distortion Corr.	On
Mode	2D
Unfiltered images	Off

Sequence - Part 1

Introduction	On
Dimension	2D
Compensate T2 decay	Off
Reduce Motion Sens.	Off
Contrasts	1
Flow comp.	No
Multi-slice mode	Interleaved
Free echo spacing	Off
Echo spacing	12.7 ms
Bandwidth	120 Hz/Px

Sequence - Part 2

Define	Turbo factor
Echo trains per slice	8
Phase correction	Automatic
Acoustic noise reduction	None
RF pulse type	Normal
Gradient mode	Normal
WARP	Off
Red. EC sensitivity	Off
Turbo factor	17

Sequence - Special

Panda Tweaks	Off
--------------	-----

Sequence - Assistant

Mode	TR
Max. TR	5000.0 ms
Allowed delay	30 s